Tax Shocks and County-Level Migration: Are the Wealthy More Mobile?

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1 Introduction

We examine whether state-level tax shocks cause high-income households to relocate to lower-tax jurisdictions.

Our primary data source is the IRS Statistics of Income (SOI) program, which publishes county-level income and migration files annually. The income files report adjusted gross income (AGI) and tax paid by county, income bracket, and year. The migration files record the number of returns, number of individuals, and cumulative AGI for every county-to-county move in each year. Our sample spans **2011–2022**.

The unit of observation in our analysis is a **county-pair × year**. So for each origin county, we observe how many people moved to each destination county in a given year, along with the income and tax characteristics of both counties.

To identify causal effects we will explore three state-level policy shocks with known effective dates:

1. **New Jersey (2018):** top marginal rate raised from 8.97% to 10.75% on income above \$5M; threshold later lowered to \$1M in 2021.
2. **New York State (2021):** top marginal rate raised from 6.85% to 9.85% on income above \$1M.
3. **Delaware (2018):** estate tax repealed (previously 16% on inheritances above \$5.49M).

Our effective tax rate variable is computed as $(Federal\ income\ tax + State\ \&\ local\ taxes\ paid)/AGI$, computed separately for high-income (AGI $>$ \$200k) and low-income (AGI $\leq$ \$200k) households in each county-year.

2 Data Construction

All the data we use is scraped from the IRS Statistics of Income (SOI) webpage. The two data series we use are county-level aggregates of household income and tax statistics during the period of 2011-2022. The IRS bins households by adjusted gross income (AGI), so information about households with AGI above $200k is separate from those with AGI below $200k (we combine all the other AGI bins and call these two groups high and low-income households). For future stages of this project, we can use this segmentation of data to observe differences in tax rates for regular people ($200k AGI is about 95th percentile) separately from differences in tax rates for very high-income people. To protect privacy, the counties where the number of tax returns in an AGI bin is fewer than 20 have all the values for that county-year-AGI bin combination suppressed to zero.

Migration data comes from a separate series of publications also from SOI. It includes the number of tax returns which were filed in one county in year 1 and another in year 2, the fips codes of those two counties, the number of people represented in the tax returns, and the total AGI of these tax returns in the second year. We construct a panel of all the county migration pair-year combinations with various income and tax statistics about the two counties. Similar to the income and tax data, if the number of tax returns in a county pair-year combination is fewer than 10, the values for that observation are suppressed to -1.

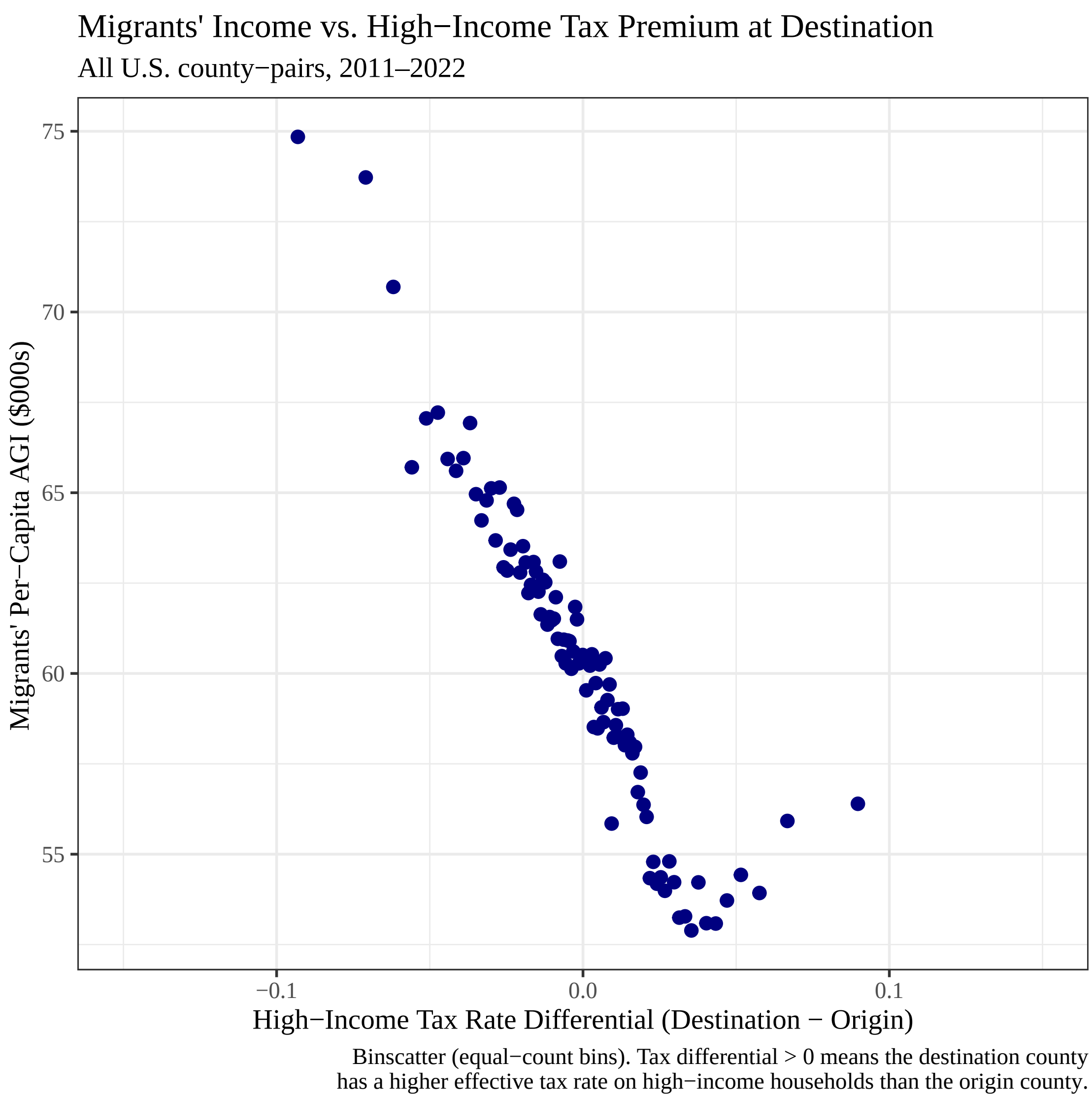
Our analysis is not so concerned with the extensive margin (a county having any migrants), so we drop all observations where migration data are suppressed to -1. These observations are entirely very small counties, and we expect this decision not to bias any estimates. In the case where a county's income data is suppressed, the effective tax rates are NaN, and those counties are omitted from regressions.

3 Descriptive Analysis

The figures below explore the unconditional relationships in our data before turning to causal identification. We ask two preliminary questions:

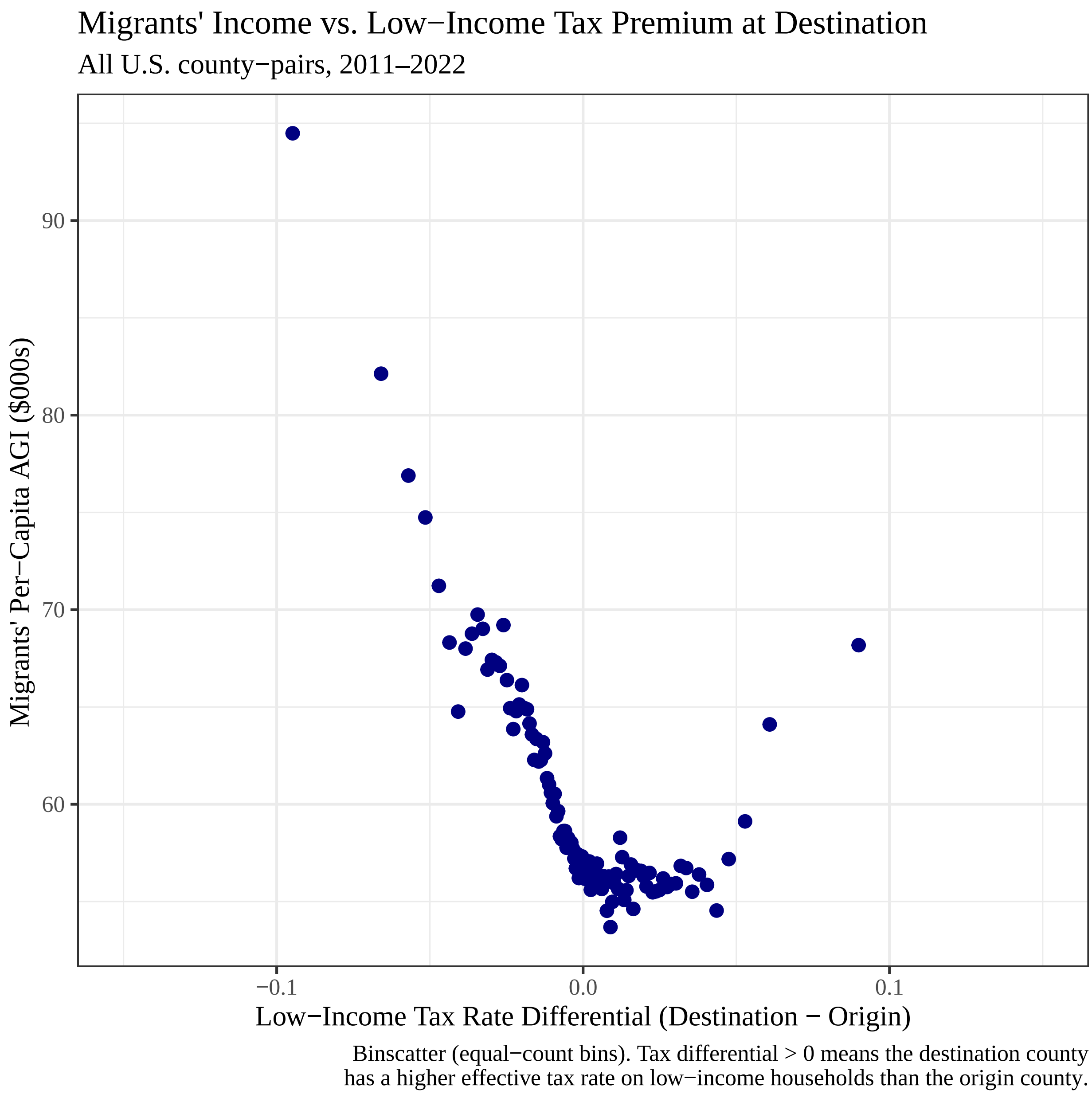
1. Across all county-pairs, do wealthier migrants tend to move toward lower-tax destinations?
2. Does this pattern differ depending on whether we measure the tax rate faced by high-income or low-income households?

3.1 Migrants' Wealth vs. High-Income Tax Differential



The downward slope indicates that wealthier migrants tend to move toward counties with lower effective tax rates on high-income households. County-pairs where the destination has a higher tax burden than the origin attract migrants with below-average incomes, while county-pairs where the destination is less taxed attract higher-income movers. This is consistent with the basic prediction of the fiscal sorting model.

3.2 Migrants' Wealth vs. Low-Income Tax Differential



The pattern here is similar to Figure 1, but the slope is shallower. Migrants' income is still negatively correlated with moving to a higher-tax destination, but the relationship is weaker when tax burden is measured using the low-income bracket. This suggests that high earners are more responsive to the tax environment at the destination than low earners are.

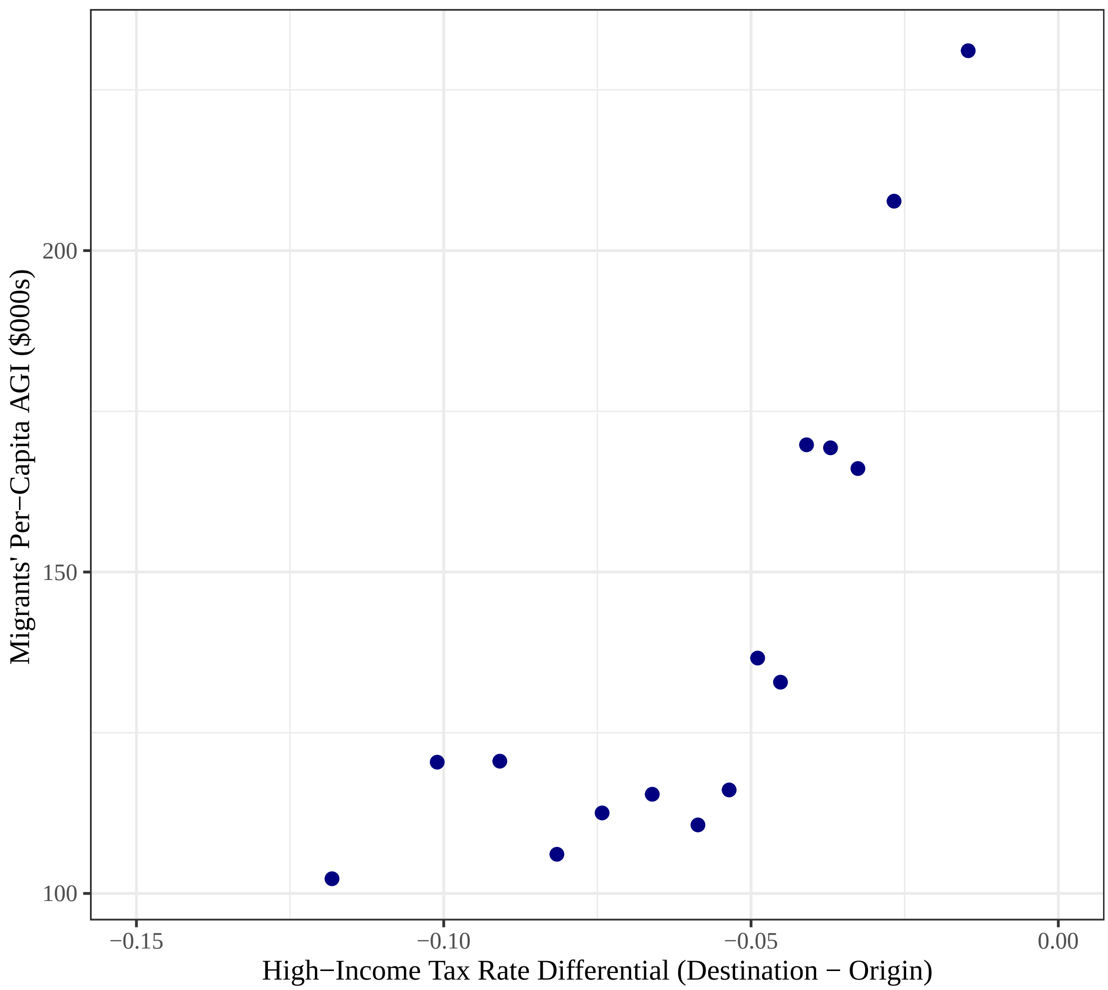
4 Manhattan Case Study

To reduce cross-sectional noise we will look **Manhattan (New York County, FIPS 36061)**, one of the highest-tax, highest-income counties in the country. We ask: among the people leaving Manhattan in any given year, does the destination county's effective high-income tax rate predict how wealthy those migrants are?

A negative slope would suggest wealthy Manhattanites preferentially exit to lower-tax destinations. A positive slope, which is what we observe, suggests something more nuanced: the wealthiest Manhattan leavers tend to relocate to other expensive, high-tax metros (San Francisco, Boston, Seattle), rather than moving purely to minimize taxes.

Manhattan Emigrants: Wealth vs. Destination Tax Rate

People leaving Manhattan (FIPS 36061), all years



Binscatter of Manhattan−origin county−pair observations.

The curve peaks in the middle of the tax differential range and falls on both ends. The upward portion on the left, where the destination is cheaper than Manhattan, actually captures the wealthiest leavers heading to places like Florida or Texas, states with lots of indemnities despite their low taxes. The fall on the right, where the destination is even more expensive than Manhattan, reflects fewer wealthy people choosing to move somewhere with a higher tax burden. Overall, the relationship is non-linear, which supports our decision to move to a log scale in the next figure.

5 Semi-causal Analysis: New York's 2021 Tax Increase

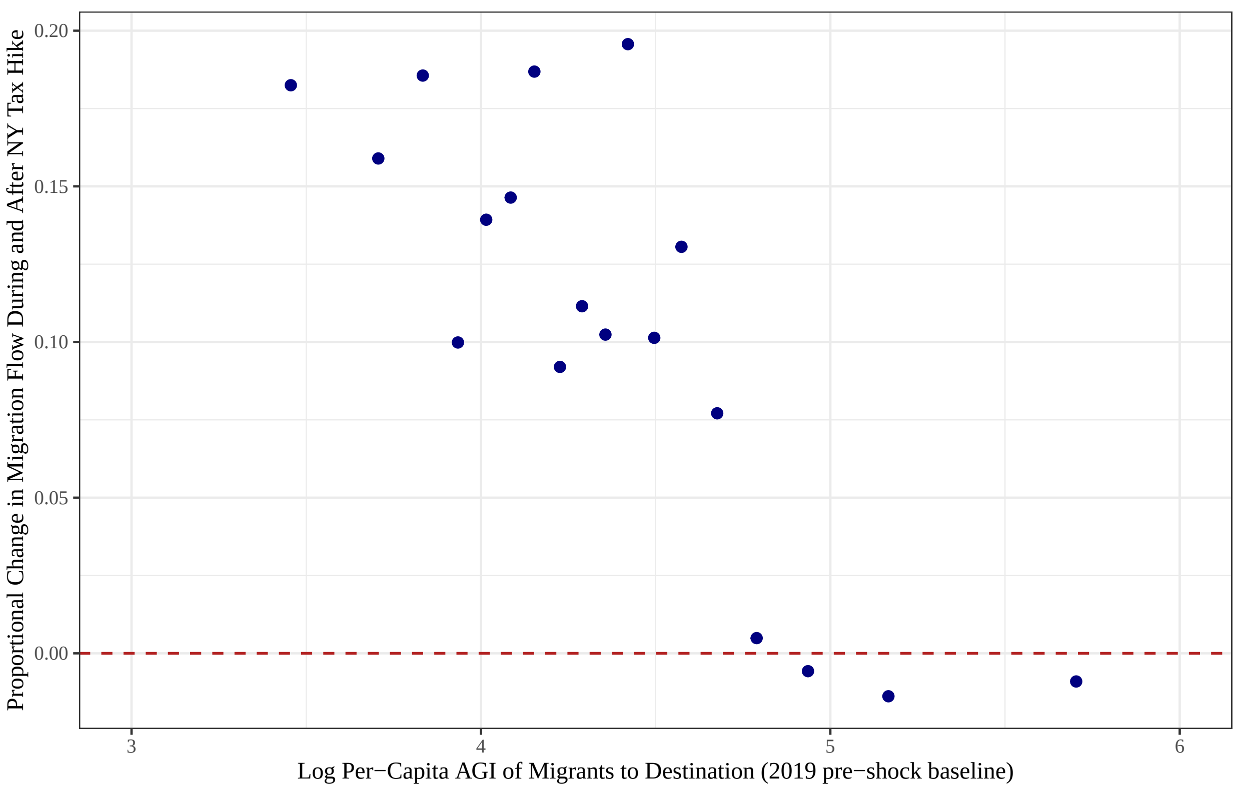
In 2021, New York State raised its top marginal income tax rate from **6.85% to 9.85%** on income exceeding $1M. Our strategy capitalizes on the fact that some destination counties outside New York were already attracting wealthy New York migrants before the shock, while others were not. If the 2021 tax increase caused a behavioral response among high earners, the counties that were drawing wealthier New Yorkers in the pre-shock period should see a larger increase in inflows after the rate rose.

The figure below plots the **proportional change** in migration flow (average of 2021-2022 moves vs. the 2020 pre-shock baseline) against the **log per-capita AGI** of people who moved from each NY county to each destination in the pre-shock year. A positive slope would support the hypothesis that high-income New Yorkers where leaving at an increased rate after the tax increase.

> **Note on year labels:** Each `year` value in the data is the *starting* year of the IRS migration file, so `year == 2019` captures people who physically moved during 2020, `year == 2020` captures moves during 2021 (the immediate shock year), and `year == 2021` captures moves during 2022. The pre-shock baseline uses `year == 2019` (moves in 2020); the post-shock period averages `year %in% c(2020, 2021)` (moves in 2021 and 2022).

NY Tax Shock: Change in Out−Migration by Destination's Previous Migrants' Income

NY−origin county−pairs; avg. post−shock flows (2021–2022) vs. pre−shock baseline (2020)



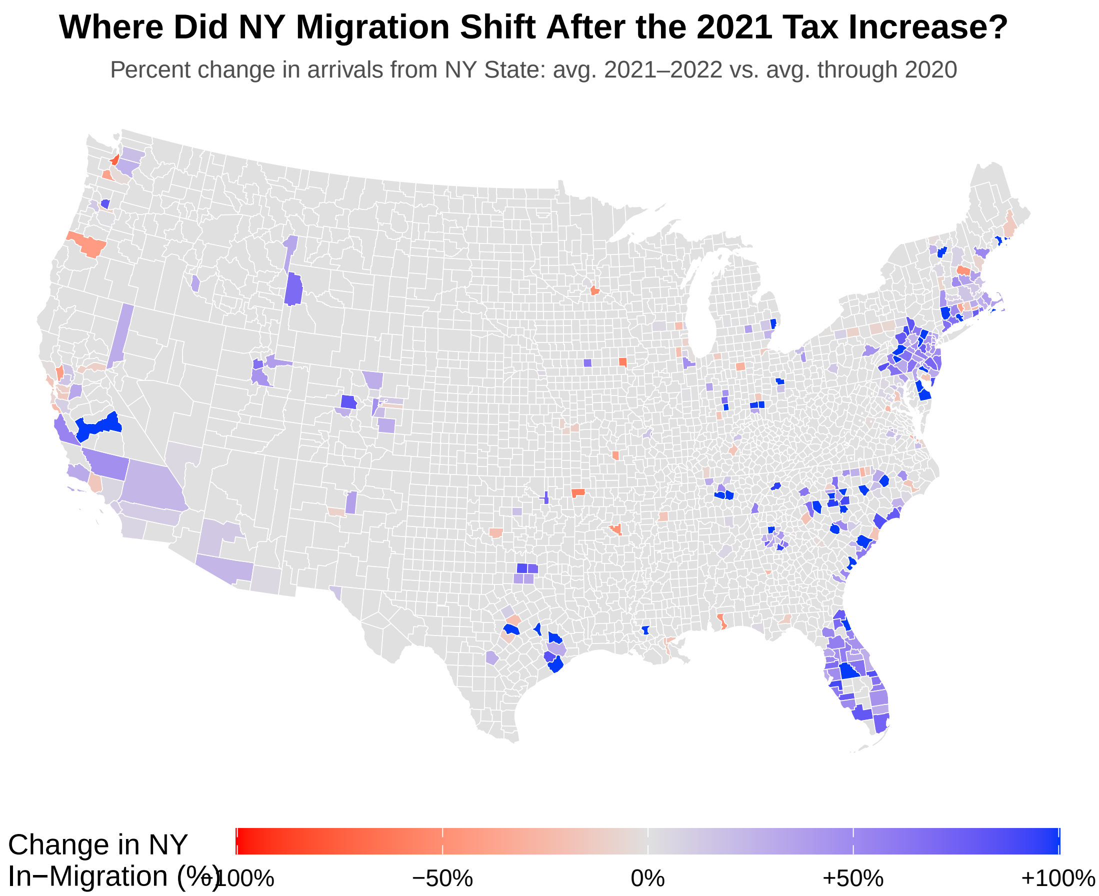
Binscatter. Y−axis: (avg. 2020–2021 label flow / 2019 label flow) − 1; zero = no change (red dashed).
X−axis: wealth of people moving from NY to each destination in the pre−shock year.
Destinations attracting wealthier NY migrants before the shock are on the right.

The slope is relatively flat and noisy, with no strong positive trend across the range of destination wealth. Destinations that were already drawing wealthier New Yorkers before the shock did not see proportionally larger inflow increases afterward. This suggests that the 2021 tax increase alone did not sharply redirect migration toward the wealthiest destinations, the behavioral response, if any, was diffused rather than concentrated among the highest-income movers.

6 Geographic Patterns: The NY 2021 Tax Shock on a Map

The binscatter in Section 4 shows the *aggregate* relationship between NY out-migration and destination wealth. The maps below show the *geographic* version of the same question: which parts of the country absorbed New York migrants, and how did that pattern shift in the year taxes went up?

6.1 Where Did NY Migrants Go? (Before vs. After 2021)



The map complements the binscatter results. Blue counties absorbed more New York migrants after the 2021 rate increase; red counties absorbed fewer. The gains are concentrated in Florida (particularly Miami-Dade and Palm Beach counties), parts of the Carolina and Texas metros, and a handful of coastal destinations. Most of the country shows little change (white), meaning the shock did not produce a broad nationwide redistribution of New York migrants. The pattern is geographically clustered rather than uniformly driven by low taxes, again pointing toward amenity sorting as a competing explanation alongside tax avoidance.

> **Note:** An animated version of this map showing Manhattan out-migration year-by-year from 2011–2021 is available in `Updated Working Files/Analysis/Map Visualization.R`. Run that script to produce `manhattan_exodus_animated.gif`. The animation makes the 2021 shift visually apparent against the prior trend.

7 Regression Analysis

For the first regression, to establish the type of analysis we intend to conduct, we estimate the following equation:

$log(n) = \alpha + \beta_1(Treat_c) + \beta_2(Post_t) + \beta_3(Treat * Post)_{c,t} + \epsilon_{c,t}$

where n is the number of tax returns moving from one county to another, $Treat_c$ is an indicator for if the county in year 1 was in New York, and $Post_t$ is an indicator for if the year of the observation was 2020 or 2021 (to capture people who moved anticipating the increase in taxes as well as people responding to the increase by leaving the following year).

We estimate the following values:

$\beta_1 = -0.0089 \quad (0.0075)$

$\beta_2 = 0.2366^{***}$ (0.0035)

$\beta_3 = 0.0677^{***}$ (0.0176)

$***$ indicates significance to 99.9% confidence.

There was a 23.7% increase in migration among counties outside of New York from the 2011-2020 average to the 2020-2022 average. The increase between the two periods for people leaving New York was only 6.8%. Despite the New York State tax increase, the number of New Yorkers fleeing the state increased by a smaller ratio than the number of people fleeing anywhere else.

So the tax increase did not produce (or happen to occur at the same time as) a massive exodus of New Yorkers. Was there any change in the average income of people leaving New York relative to people leaving anywhere else?

For this, we estimate the following regression:

$log(\frac{AGI}{n}) = \alpha + \beta_1(Treat_c) + \beta_2(Post_t) + \beta_3(Treat * Post)_{c,t} + \epsilon_{c,t}$

The only difference is that number of returns has been replaced by mean AGI of those tax returns. We estimate the following coefficients:

$\beta_1 = 0.1871^{***}$ (0.0036)

$\beta_2 = 0.2568^{***}$ (0.0017)

$\beta_3 = 0.0693^{***}$ (0.0085)

The average income of people moving out of New York State during 2011-2022 was 19% higher than the people leaving anywhere else. There was a 26% increase in mean AGI of people leaving a county outside of NY from the 2011-2020 average to the 2020-2022 average. The increase in mean AGI of people leaving NY from the 2011-2020 average to the 2020-2022 average was only 7%. This corroborates the estimates in the first estimating equation, namely that the well-monied New Yorkers did not obviously flee the 2021 tax hike.

8 Reproducibility

This document was produced using **R** and **RMarkdown**. To reproduce it on any machine:

1. **Clone the repository** from GitHub.
2. **Open the project** by double-clicking `Data-Analysis-Project.Rproj` in RStudio. This sets the working directory to the project root automatically — do not open the `.Rmd` file directly from File Explorer/Finder, as paths will break.
3. **Knit the document**: open `Project Write-Ups/Stage_Two.Rmd` and click *Knit*, or run:

   ```
   rmarkdown::render("Project Write-Ups/Stage_Two.Rmd")
   ```

4. **Packages are installed automatically** the first time you knit. No manual `install.packages()` calls are needed.
5. **The cleaned data file** (`Updated Working Files/Clean Data/cleanMigrationData.RData`) is committed to the repository. As long as this file is present, no scraping or data-cleaning scripts need to be re-run.

   Note on the map section (Section 5): The geographic figure downloads U.S. county shapefiles from the U.S. Census Bureau via the `tigris` package. This requires an internet connection on the first knit only; all subsequent knits use a local cache.

If `cleanMigrationData.RData` has been deleted, run the scripts in this order before knitting:

```
# Step 1 - only needed if Temp Data/ files were also deleted (requires internet)
source("Data Wrangling/Data Scraping.R")

# Step 2 - regenerates cleanMigrationData.RData from Temp Data/ files
source("Data Wrangling/Cleaning and Merging.R")
```

Session info at time of submission:

```
sessionInfo()
```

```
## R version 4.5.1 (2025-06-13 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.0
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: America/Denver
## tzcode source: internal
```

```
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] here_1.0.2      scales_1.4.0    sf_1.0-23       tigris_2.2.1
##  [5] fixest_0.13.2   binsreg_1.1     lubridate_1.9.4 forcats_1.0.1
##  [9] stringr_1.5.2   dplyr_1.1.4     purrr_1.2.1     readr_2.1.6
## [13] tidyr_1.3.1     tibble_3.3.0    ggplot2_4.0.0   tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] gtable_0.3.6        xfun_0.53           lattice_0.22-7
##  [4] tzdb_0.5.0          numDeriv_2016.8-1.1 vctrs_0.7.2
##  [7] tools_4.5.1         generics_0.1.4      sandwich_3.1-1
## [10] proxy_0.4-27        pkgconfig_2.0.3     Matrix_1.7-3
## [13] KernSmooth_2.23-26  RColorBrewer_1.1-3  S7_0.2.0
## [16] stringmagic_1.2.0   uuid_1.2-2          lifecycle_1.0.4
## [19] compiler_4.5.1      farver_2.1.2        MatrixModels_0.5-4
## [22] SparseM_1.84-2      quantreg_6.1        htmltools_0.5.8.1
## [25] class_7.3-23        yaml_2.3.10         Formula_1.2-5
## [28] pillar_1.11.1       MASS_7.3-65         classInt_0.4-11
## [31] nlme_3.1-168        tidyselect_1.2.1    digest_0.6.37
## [34] stringi_1.8.7       splines_4.5.1       rprojroot_2.1.1
## [37] fastmap_1.2.0       grid_4.5.1          cli_3.6.5
## [40] magrittr_2.0.4      survival_3.8-3      e1071_1.7-16
## [43] withr_3.0.2         dreamerr_1.5.0      rappdirs_0.3.3
## [46] timechange_0.3.0    rmarkdown_2.30      httr_1.4.7
## [49] matrixStats_1.5.0   hms_1.1.4           zoo_1.8-14
## [52] evaluate_1.0.5      knitr_1.50          rlang_1.1.7
## [55] Rcpp_1.1.0          glue_1.8.0          DBI_1.2.3
## [58] rstudioapi_0.18.0   R6_2.6.1            units_1.0-0
```